From Scientific Evidence to Decision Intelligence

Scientific principles • Business value • Applications • Future integrations

CLIENT OVERVIEW | OCTOBER 2026

A researcher-informed, AI-assisted approach to making complex scientific information easier to understand, compare, and use.

Executive overview

Scientific and pharmaceutical organizations face a familiar problem: information is abundant, but finding the evidence that answers a particular question takes time. The Scientific Intelligence Platform is a developing AI-assisted prototype designed to organize selected scientific documents, preserve essential evidence and context, and support question-driven scientific and business interpretation.

The aim is not simply to shorten papers. It is to identify what matters, retain the basis for scientific claims, and communicate findings in forms useful to researchers, R&D leaders, and other decision-makers.

01 Scientific foundations: why we select this information

The platform's analytical design draws on the founder's MSc training in Applied Neuroscience, extensive experience reviewing complex scientific literature, and professional work in the biomedical sector. Its information categories reflect how researchers establish what a study investigated, how it was performed, what it found, and what can reasonably be concluded.

Researcher-informed, staged scientific reading

The academic workflow begins with abstract-based relevance assessment, moves to selective reading of research questions, methods, results, discussion and conclusions, and reserves full critical examination for particularly important studies. It is conceptually aligned with Keshav's Three-Pass Approach (2007), although the workflow is not identical.

The current prototype analyzes documents already selected by the user; it does not automatically perform the complete three-stage literature selection process.

Scientific context before conclusions

- Research objectives and the question being investigated
- Study design, experimental models or populations, methods and outcome measures
- Findings and relevant quantitative results, including experimental conditions
- Conclusions, limitations and uncertainty
- Distinction between observed evidence, scientific interpretation and possible applications

The approach is intended to cover relevant molecular, clinical, neuroscientific, psychiatric, psychological and behavioral evidence without assuming every finding describes a biological mechanism. Broader cross-disciplinary coverage remains an ongoing refinement.

01A Information collected and presented

The report is designed for progressive reading: a rapid orientation to the study, followed by structured scientific detail and, where relevant, critical interpretation and decision implications. This presentation is informed by the Three-Pass Approach (Keshav, 2007). It is a reporting design principle; the current prototype does not autonomously execute all three reading passes.

First view — Study orientation

A reader should quickly understand the study category and context, the research question, the main contribution, and the scope of its claims. Keshav's Five Cs—Category, Context, Correctness, Contributions and Clarity—inform this first view. Correctness is treated as a reasoned methodological consideration rather than a simple pass/fail score; clarity is also a standard for the report itself.

Second view — Scientific evidence

The report should present the original research question together with a clear explanation; study design; all relevant methods, models, populations and sample details; measured outcomes; distinct scientific findings with quantitative results where available; conclusions; and source references. Scientific findings may be molecular, clinical, neuroscientific, psychological, behavioral or otherwise relevant to the study. No arbitrary cap should suppress distinct meaningful methods or findings.

Third view — Critical interpretation

Where supported, present methodological limitations, uncertainty, alternative explanations, differences between experimental conditions, genuine cross-study contradictions, knowledge gaps, and implications for further research. Distinguish what authors reported from AI-generated synthesis and potential business implications. Avoid presenting association as causation or preclinical results as clinical evidence.

Authors, institutions and research expertise

Extract author names, corresponding authors, senior authors and institutional affiliations as reported in the paper. Identify a principal investigator only where the source explicitly establishes that role; first or last authorship alone does not prove PI status. An optional Researcher Expertise Profile may enrich the report with externally verified research areas, laboratories, ORCID/institutional profiles and relevant publications. External enrichment requires source links, identity matching and a clear "not verified" status where evidence is insufficient.

Presentation rules

Use descriptive headings and separate study facts, interpretations and proposed actions. Provide a quick overview followed by expandable depth in the reporting design; explain technical terms without discarding precise terminology; show findings in readable numbered units; preserve context, caveats and evidence identifiers; and label missing or unreported information explicitly rather than guessing.

Suggested report headings: Study Overview; Research Question — Explained; Study Type & Context; Authors & Research Institutions; Methods & Study Design; Key Scientific Findings; Quantitative Results; Plain-Language Interpretation; Conclusions & Limitations; Cross-Study Comparison (when applicable); Scientific & Pharmaceutical Implications; Sources. Researcher Expertise Profile is optional and separately verified.

02 Information quality: how we present evidence

The design is informed by Wang and Strong's research on data quality beyond accuracy (1996). We express our output goals through three complementary principles:

Principle	Meaning	Implication for reporting
Fidelity	Preserve scientific meaning and source context	Keep methods, qualifications, uncertainty and traceability
Utility	Answer the reader's actual question	Make findings understandable and decision-relevant
Density	Reduce unnecessary words, not useful findings	Remove repetition without arbitrary limits on distinct evidence

Our guiding principle: Preserve information. Maximize understanding. Minimize unnecessary words.

A scientifically accurate finding can still be misleading if detached from its model, population, measurement or limitations. Equally, a concise report is not necessarily a useful one if it omits meaningful findings.

03 What organizations can gain

Value	Potential benefit
Faster access to evidence	Less repetitive reading and manual extraction of selected documents
Question-driven answers	Different users can explore the same evidence base for different purposes
Signal within information overload	Identify meaningful findings without losing necessary context
More consistent comparisons	Organize methods, findings and limitations across studies
Reusable knowledge	Use structured evidence for follow-up questions and reporting
Clearer communication	Translate technical findings for multidisciplinary scientific and management audiences

These are intended benefits, not measured performance claims. Actual time savings and accuracy improvements require evaluation in the client's workflow.

04 Example applications in pharmaceutical and scientific work

Area	Illustrative questions and uses
Discovery research	What pathways, targets or mechanisms have been investigated?
Preclinical and translational R&D	How do findings differ across experimental models and conditions?

Clinical research	Which populations, outcomes and limitations were reported?
R&D prioritization	What uncertainties or evidence gaps warrant further investigation?
Scientific competitive intelligence	How do selected published approaches and findings compare?
Business development	Which scientific observations may be relevant to an opportunity—and what remains unproven?
Scientific communication	How can complex evidence be explained accurately to a defined audience?

One evidence base, different questions

A molecular scientist may investigate resistance mechanisms, while an R&D director asks which uncertainties could affect the next development step. A business development team may ask which findings justify further diligence. The system is intended to support these different perspectives without treating strategic inferences as established scientific facts.

05 Evidence, interpretation and decisions

Layer	Question answered
Scientific evidence	What did the source documents actually report?
Scientific interpretation	What do the findings mean, and how do they relate?
Decision support	What might be worth investigating or considering next?

The distinction matters: laboratory findings do not automatically demonstrate clinical effectiveness, and scientific plausibility does not establish commercial feasibility. The approach is informed by principles of evidence appraisal and Evidence-to-Decision thinking, but the platform is not a formal GRADE assessment, systematic review or meta-analysis.

06 Current prototype capabilities

- Analysis of selected scientific PDF documents
- Structured scientific information extraction
- AI-assisted scientific and pharmaceutical interpretation
- Questions across selected documents
- Structured outputs, including Excel reporting

The prototype has been explored with selected biomedical papers. Its results require scientific review, and it has not completed independent, system-wide validation. Further work focuses on clearer explanations, broader evidence types and integrated evidence-linked reporting.

07 Potential API integrations and follow-on projects

The platform can provide a foundation for client-specific extensions. Subject to data access, licensing, security and technical feasibility, future projects could connect it to research and molecular databases or authorized internal systems.

Potential connection	Possible purpose

PubMed / Europe PMC / Crossref	Literature discovery, metadata and accessible research content
UniProt / Ensembl / RCSB PDB	Protein, gene and structural biology context
ChEMBL / PubChem	Compounds, bioactivity and chemical information
ClinicalTrials.gov	Registered clinical studies and trial information
Internal research repositories	Authorized company reports, experiments and document collections

Examples of tailored follow-on work include automated research searches, target and compound intelligence, program-specific scientific assistants, recurring evidence updates, custom R&D dashboards and audience-specific translational writing. These are development opportunities, not current built-in integrations.

08 Responsible use and scientific oversight

The system is intended to augment expert judgment, not replace it. Important conclusions should remain source-linked, uncertainties should be explicit, and consequential scientific or business decisions require human validation. The design direction is consistent with broad principles of the NIST AI Risk Management Framework; no formal NIST certification or compliance is claimed.

Conclusion

The platform connects a researcher-informed approach to reading scientific literature with structured evidence organization, information quality principles and flexible, question-driven analysis. Its value lies in helping organizations move from information overload to scientifically grounded understanding—and, ultimately, better-informed questions and decisions.

Scientific information should become easier to use without becoming less scientifically reliable.

Selected methodological references

Keshav, S. (2007). How to Read a Paper. ACM SIGCOMM Computer Communication Review, 37(3), 83–84. doi:10.1145/1273445.1273458

Wang, R. Y., & Strong, D. M. (1996). Beyond Accuracy: What Data Quality Means to Data Consumers. Journal of Management Information Systems, 12(4), 5–33. doi:10.1080/07421222.1996.11518099

GRADE Working Group. Evidence to Decision frameworks. gradeworkinggroup.org

NIST (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). doi:10.6028/NIST.AI.100-1